

AKHILESH KUMAR MISHRA

Python Full Stack Developer

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PROFILE

Software Development Engineer specializing in Python Full Stack development, with a track record of building secure, scalable backend architectures and web systems. Expert in Python, Django, Django REST Framework (DRF), FastAPI, Celery, and database design in PostgreSQL/MySQL. Proven experience in migrating monoliths to async microservices, designing event-driven pipelines using Celery/Redis, and building real-time dashboards via WebSockets. Optimizes database queries and scales backend infrastructures using Docker.

CORE TECHNOLOGY

Backend: Python, Django, FastAPI, Celery, Django REST Framework, SQLAlchemy (asyncio), Pandas

Frontend & APIs: React.js, JavaScript, HTML5, CSS3, Bootstrap, RESTful APIs, WebSockets, JWT Security

Data & Messaging: PostgreSQL, MySQL, SQLite, Redis cache/broker, database schema design, query indexing

Tools & Cloud: Git, GitHub, Docker, Postman, Linux environments, OpenCV, LBPH face recognition

EXPERIENCE

Software Development Engineer — CSC e-Governance Services India Ltd. (MeitY, Government of India)

Ministry of Electronics & Information Technology, Govt. of India

12/2023 – PRESENT | New Delhi, India

- Re-architected monolithic web services into scalable, asynchronous **FastAPI** and **Django** microservices, improving reliability for **2 million+ users**.
- Designed and scaled transaction orchestration services for AEPS, MATM, and VATM platforms using Django REST Framework, handling **500,000+ daily transactions**.
- Implemented secure API authentication using **JWT** and role-based access control (**RBAC**), including OTP verification, reducing authentication latency by 50%.
- Developed an asynchronous task queuing pipeline using **Celery** and **Redis**, reducing processing bottlenecks by 35% under peak traffic.
- Tuned **PostgreSQL** and **MySQL** database schemas using indexing and query partitioning, lowering transaction query latency by 45%.
- Constructed real-time transaction tracking APIs utilizing **FastAPI WebSockets** to stream live status updates in under 500ms.

Freelance Software Engineer — Workforce Telemetry & Operations

Real-Time Telemetry & Geofencing Platform

06/2023 – 11/2023 | Remote, India

- Developed a real-time workforce telemetry and geofencing system utilizing **FastAPI**, **SQLAlchemy** (asyncio), and **React.js**, serving **1,000+ active field agents**.
- Built an interactive live telemetry dashboard in **React** using Ant Design to visualize coordinates and speed, reducing UI load times by 30%.
- Implemented a high-performance, asynchronous ingestion API to capture GPS locations, processing **10,000+ concurrent GPS pings with 99.9% uptime**.

SELECTED PROJECTS

Facial Recognition Attendance Dashboard

Python / OpenCV / Tkinter / MySQL / Pandas

Developed an automated student attendance tracking application in Python using OpenCV (Haar Cascade & LBPH) and MySQL, with custom Excel reports via Pandas.

Himalayan Edges Website Development

Python / Django / React.js / SQLite / AWS / PWA

Designed and launched an e-commerce platform using Django and React.js with PWA features, hosted on AWS, boosting page speeds and usage by 30%.

Persistent Task Management Dashboard

Python / Tkinter / JSON / MVC Patterns

Programmed a desktop utility for task planning, category filtering, and JSON data persistence, featuring clean MVC architecture patterns.

Traveling Portal & Landing Page

HTML5 / CSS3 / JavaScript / Bootstrap

A fully responsive, modern front-end landing page for a travel agency built with HTML, CSS, and Bootstrap layouts.

EDUCATION

Bachelor of Technology (B.Tech.) | Kanpur Institute of Technology, Kanpur, UP

2019 - 2023

CERTIFICATIONS

Advanced MERN Stack Developer Certification
Database Systems & SQL Certification
Data Structures & Algorithms Certification
IIT Kanpur Cyber Security Certification

ACHIEVEMENTS

- Graduated with First Class Honors, maintaining top academic rank.
- Earned a 4-Star Gold Badge in SQL on HackerRank.